LOH WEI JIAN

3rd Year Bachelor of Mechanical Engineering Student
22 Years Old
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Internship Period: 20 July - 27 Sep 2026

EDUCATION

UNIVERSITY OF MALAYA — *Expected 2027*

Bachelor of Mechanical Engineering
CGPA 3.51

- Proficient in **SolidWorks** for 3D modeling and **ANSYS** for Finite Element Analysis, conducting structural and thermal simulations to validate design integrity.
- Expertise in Static and Dynamic loading analysis, including bridge, beam stress, force and moment analysis. Advanced knowledge of material properties and specialized materials, including **Ceramics** (Silicon).
- Developed algorithmic solutions and engineering tools using **C++** and **Python** for complex problem-solving.
- Skilled in designing integrated systems using **Arduino IDE**, **EasyEDA and TinkerCAD**, competent in circuit design, sensor instrumentation, and basic electronic component integration.
- Applied **MATLAB** for numerical computing and system simulation, able to utilise **LabVIEW** for mechanical vibration analysis and automated data acquisition, and **Microsoft Excel** for data record and analysis.

WORK EXPERIENCE

FREELANCE — *Oct 2024 - Present*

Photographer

- Directed and guided models and subjects through clear, **effective communication** to achieve specific visual objectives, demonstrating the ability to **lead** and **coordinate** in high-pressure environments.

KUMON — *July - Oct 2023*

English, Advanced/Modern Mathematics Tutor

- Facilitated **English language** literacy for primary students and taught **Modern and Advanced Mathematics** for a diverse cohort ranging from primary to secondary school levels.
- Demonstrated high adaptability of different communication methods to suit various age groups, ensuring student understanding of complex mathematical concepts.

ACTIVITIES

IMECHE UM STUDENT CHAPTER — *July 2025 - Present*

Treasurer

- Orchestrated the fiscal management of the organization, tracking cash flow and departmental budgets for various large-scale student events.
- Able to streamline the reimbursement process between the student body and the Faculty Administration.

ENGINEERING SOCIETY OF UM
UM TECHNOTHON
CROSSTALK PERFORMANCE COMMITTEE (CHINESE SOCIETY)
IMECHE UM STUDENT CHAPTER *July 2024 - July 2025*

Head of Departments

- Simultaneously spearheaded **four distinct departments**.
- Directed the full-stack design and maintenance of organizational websites, prioritizing **UI/UX optimization** and responsive layouts to enhance user engagement and information accessibility.
- Successfully navigated the **high-pressure environment** of managing four concurrent HOD roles while maintaining academic rigor within the Mechanical Engineering curriculum.

UNIVERSITY PROJECTS

ECO FRIENDLY BARNACLE CLEANER *Ongoing*

SDG 14 Life Below Water

- Conducted comprehensive user-requirement surveys and stakeholder analysis to identify pain points in traditional hull maintenance.
- Developed high-fidelity 3D models and assembly drawing based on decision-making matrices for multiply concept selections.

DESIGN ON SPUR GEAR SPEED REDUCER *Jun 2025*

- Engineered a **speed reducer** with a 1:2 gear ratio.
- Able to calculate critical parameters, analysing von Mises stress, resultant displacement, factor of safety, and structural study via simulation.

ICE-CAP MAPPING DRONES *Jan 2025*

SDG 13 Climate Action

- Conducted a comprehensive teardown of a drone system to analyze electronic architecture
- Utilized **Tinkercad** to simulate and validate circuit behavior including transistors, pull-down resistors, and decoupling capacitors.
- Developed and tested **Arduino-based logic** to control motor states and LED indicators, utilizing **PWM** for speed control and digital signal processing for switch-state monitoring.

MOSQUITO TRAPPER: MOSQUARIUM *Jan 2025*

- Designed the integration of a **Blue UV light attraction system** and an electric neutralizer grid, balancing low energy consumption (9V/4.5W) with high operational efficiency.
- Able to streamline the reimbursement process between the student body and the Faculty Administration.

ADDITIONAL

Technical Skills: Proficient in Solidworks, Ansys, Labview, Adobe, Arduino IDE, EasyEDA, TinkerCAD, Microsoft, C++, Python, Matlab.
Languages: Proficienct in English, Malay, Chinese